

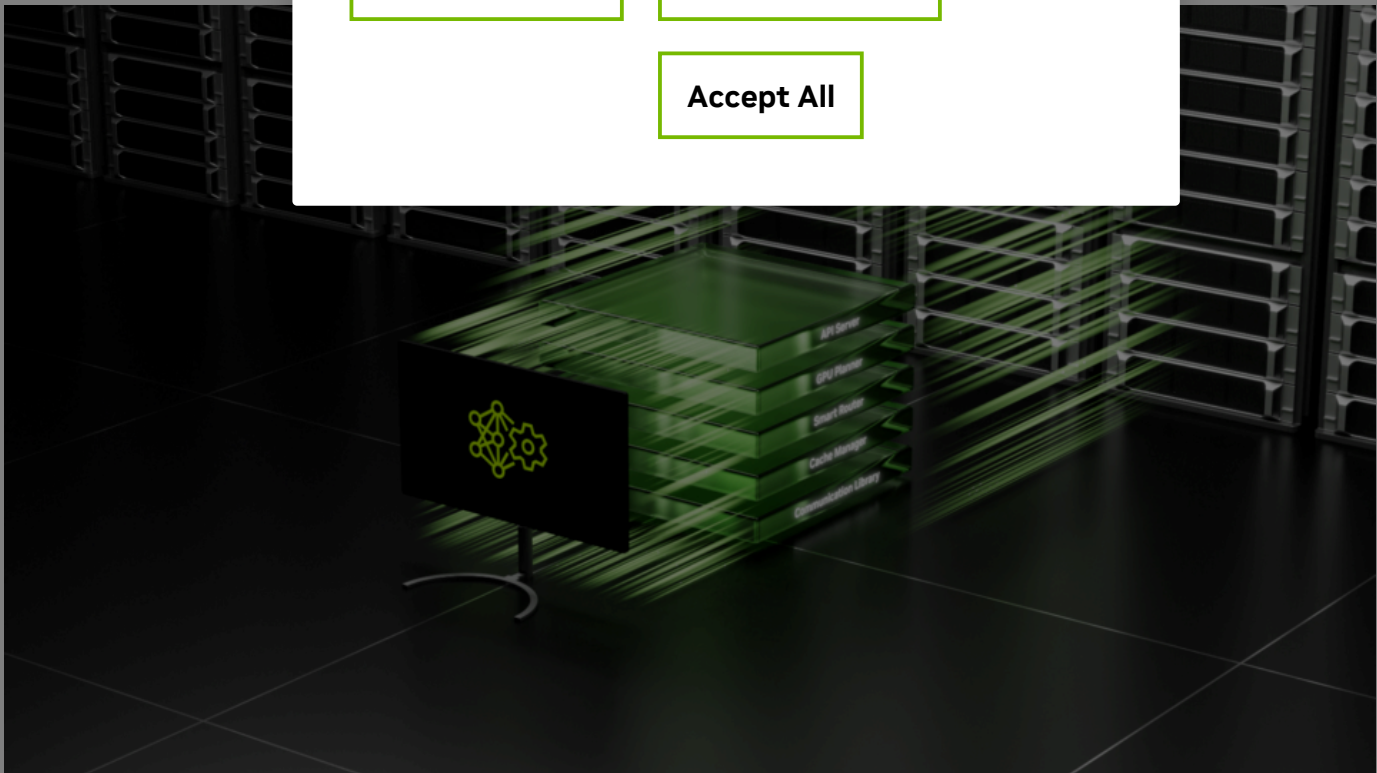
NVIDIA Latency Framework Reasons

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Mar 18, 2025

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By [Amr Elmeleegy](#), [Harry Kim](#), [David Zier](#), [Kyle Kranen](#), [Neelay Shah](#), [Ryan Olson](#) and [Omri Kahalon](#)



AI-Generated Summary



and reasoning models in large-scale distributed environments. The framework boosts the number of requests served by up to 30x when running the open-source DeepSeek-R1 models on NVIDIA Blackwell. It integrates with PyTorch, SGLang, [NVIDIA TensorRT-LLM](#), and other tools that empower developers to build high-performance AI applications.

NVIDIA Dynamo introduces

- Disaggregated pre-training and inference
- Dynamic scheduling and resource allocation
- LLM-aware request scheduling
- Accelerated asynchronous inference
- KV cache offloading

Starting today, NVIDIA is open-sourcing the NVIDIA Dynamo repo. For enterprises, we offer a commercial support, and stability SLA. Learn more about NVIDIA AI Enterprise [here](#).

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This post explains the architecture and key components of NVIDIA Dynamo, highlighting how they facilitate cost-effective disaggregated serving and scaling of generative AI models, from a single GPU to thousands of GPUs.

Interview with NVIDIA Dynamo Architect Kyle Kranen

NVIDIA Developer

Accelerating AI inference in multinode deployment

AI inference will help reasoning models in more intuitive ways. challenges for those for AI.

When NVIDIA first in AI innovation and drive server that consolidate PyTorch, ONNX, Open inference costs and

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Triton has been downloaded over 1 million times from [NVIDIA NGC](#) and is today used by some of the world's leading organizations to deploy AI models in production, including [Amazon](#), [Microsoft](#), [Oracle Cloud](#), [DocuSign](#), [Perplexity](#), [Snap](#), and many others.

Since the launch of Triton, open-source model sizes have grown dramatically—by almost 2,000x—and are now increasingly integrated into [agentic AI](#) workflows that require interaction with multiple other models. Deploying these models and workflows in production environments involves distributing them across multiple nodes, which demands careful orchestration and coordination across large fleets of GPUs. The complexity intensifies with the introduction of new distributed inference optimization methods, such as disaggregated serving, that split the response to a single user request across different GPUs. This makes collaboration and efficient data transfer between them even more challenging.

To tackle the challenges of distributed generative AI inference serving, we are releasing NVIDIA Dynamo. Dynamo builds on the successes of Triton with a new modular architecture designed to serve generative AI models in multinode distributed environments.

NVIDIA Dynamo enables seamless scaling of inference workloads across GPU nodes and dynamic GPU worker allocation to efficiently respond to fluctuating user demand and handle

LLM inference serving optimization techniques such as disaggregated serving, which separates the different phases of inference onto distinct GPU devices to boost inference performance.

Boosting NVIDIA G

Traditional LLM deployment on a single GPU or node, despite its simplicity, has hindered performance due to inefficient resource utilization.

The prefill phase processes the entire input sequence, while the decode phase generates tokens sequentially. Running these phases on the same GPU device for long input sequences. Additionally, the sequential nature of these phases reduces flexibility, causing model underutilization.

To address these issues, disaggregated serving separates the prefill and decode phases onto different GPUs or nodes. This enables developers to optimize each phase independently, applying different model parallelism strategies and assigning different GPU devices to each phase (Figure 1).

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on a single GPU device, which hinders performance due to inefficient resource utilization.

compute-bound, leading to inefficient resource utilization. Disaggregating these phases onto different GPU devices for long input sequences improves flexibility and resource utilization.

For instance, low tensor parallelism can be used in the prefill phase to reduce communication overhead, while high tensor parallelism can improve memory operations in the decode phase.

This approach enables better control of inference costs, and

provides better control of token latency

(ITL).

When serving the op

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Dynamo more than c

NVIDIA Hopper.

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NVIDIA Dynamo,

x. NVIDIA

70B model on

Figure 2. NVIDIA Dynamo delivers exceptional throughput performance when running DeepSeek-R1 671B model on NVIDIA GB200 NVL72 boosting performance by 30x. It more than doubles performance on the Llama 70B model running on NVIDIA Hopper GPUs

Left: TensorRT-LLM, FP4, ISL/OSL: 32K/8K. Without Dynamo: Inflight Batching, TEP16PP4DP4. With Dynamo:

Disaggregated Serving, Context: EP4DP16, Generation: EP64DP3. Projected performance subject to change. Right:

vLLM, FP8, ISL/OSL: 3K/50. Without Dynamo: Inflight Batching, TP8DP2. With Dynamo: Disaggregated Serving,

Context: TP2DP4, Generation: TP8.

To enable large-scale distributed and disaggregated inference serving, NVIDIA Dynamo includes four key innovations:

NVIDIA Dynamo Distributed RV Cache Manager

- NVIDIA Inference Transfer Library (NIXL)

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Figure 3. NVIDIA Dynamo architecture

NVIDIA Dynamo Planner: Optimizing GPU resources for distributed inference

In large-scale distributed and disaggregated-serving inference systems, efficiently managing GPU resources is essential to maximize throughput and minimize latency. While disaggregated serving can significantly enhance inference throughput and efficiency, it might not always be the most effective solution for every incoming request.

Imagine a scenario where a surge of summarization requests with long input sequence lengths (ISLs) but short output sequence lengths (OSLs) overwhelms the prefill GPUs. While the decode GPUs remain underutilized, the prefill GPUs become a bottleneck. In this case, allowing decode GPUs to perform both prefill and decode tasks in a traditional aggregated manner, or shifting

Deciding between disaggregated and aggregated serving or how many GPUs should be assigned to each phase requires a complex analysis of the system state. The time needed to transfer data between GPUs, the size of the queue wait times for incoming requests, and the current state of the system aggregated and disaggregated configurations. These decisions can quickly become complex.

This is where NVIDIA's GPU Planner comes in. It analyzes GPU capacity metrics in order to make the optimal decision on how to serve SLOs such as TTFT and TTP. It can adapt to changing requests with or without disaggregation. NVIDIA's Dynamo Planner ensures that the system is adapting to fluctuating

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Figure 4. GPU Planner analyzes GPU capacity metrics to make the optimal decision on how to serve incoming requests or allocate GPU workers

NVIDIA Dynamo Smart Router: Reducing costly recomputation of KV cache

the size of the input request. Reusing KV cache avoids the need to recompute it from scratch, reducing inference time and compute resources. This is particularly advantageous in use cases where the same request is repeated, such as in chatbot interactions or multiturn chatbot interactions. The NVIDIA Dynamo Smart Router is required to check whether the KV cache is valid for the current request.

The NVIDIA Dynamo Smart Router disaggregates deployment across multiple nodes, reducing their costly recomputations. This allows the tracking of KV cache across nodes, which utilizes specialized algorithms to ensure relevant blocks are reused across nodes. The NVIDIA Dynamo Smart Router also enables the need for a KV cache tree, which is a tree structure that stores KV cache blocks. It also allows for the most efficient reuse of KV cache blocks across nodes.

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Figure 5. NVIDIA Dynamo Smart Router avoids KV cache recomputations accelerating model response time and enhancing user experience

2x HGX-H100 nodes. 8x DeepSeek-R1-Distill-Llama-70B. vLLM, FP8, Tensor Parallel: 2

Data Source: 100K real R1 requests, Avg ISL/OSL: 4K/800

When new inference requests arrive, the NVIDIA Dynamo Smart Router computes an overlap score between incoming requests and the KV cache blocks already active across all the memory of all GPUs in the distributed cluster. By factoring in the overlap score and the distribution of

Unlike round-robin or load-based routing, this approach optimizes overall system performance by factoring in cache reuse. By intelligently routing requests to the cache that best handles the request, NVIDIA Dynamo KV Cache Manager can significantly reduce the cost of KV cache, NVIDIA Dynamo KV Cache Manager can significantly reduce the cost of KV cache to respond to more requests. This is a significant benefit for organizations with high-volume investments.

NVIDIA D Manager effective

Calculating KV cache reuse is a complex task. As AI demand increases, the volume of KV cache data grows rapidly, and managing it can quickly become prohibitively costly. This creates a significant challenge for AI inference teams trying to efficiently manage KV cache reuse without exceeding their budget.

The NVIDIA Dynamo KV Cache Manager feature addresses this challenge by enabling the offloading of older or less frequently accessed KV cache blocks to more cost-effective memory and storage solutions, such as CPU host memory, local storage or networked object storage. This capability enables organizations to store up to petabytes of KV cache data at a fraction of the cost of keeping it in GPU memory. By offloading KV cache to alternative memory hierarchies, developers can free up valuable GPU resources while still retaining and reusing historical KV cache to reduce inference computation costs.

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Figure 6. NVIDIA Dynamo KV Cache Manager

The NVIDIA Dynamo KV Cache Manager uses advanced caching policies that prioritize placing frequently accessed data in GPU memory, while less accessed data is moved to shared CPU host memory, SSDs, or networked object storage. It incorporates intelligent eviction policies that strike a balance between over-caching (which can introduce lookup latencies) and under-caching (which leads to missed lookups and KV cache recomputation).

Additionally, this feature can manage KV cache across multiple GPU nodes, supporting both distributed and disaggregated inference serving, and offers hierarchical caching capabilities, creating offloading strategies at the GPU, node, and cluster levels.

The NVIDIA Dynamo KV Cache Manager is designed to be framework-agnostic to support various backends, including PyTorch, SGLang, TensorRT-LLM, and vLLM, and to facilitate the scaling of KV cache storage across large, distributed clusters using [NVIDIA NVLink](#), [NVIDIA Quantum](#) switches, and [NVIDIA Spectrum](#) switches.

(NIXL): Low-latency, hardware-agnostic communication

Large-scale distributed training pipeline, and expert throughput communication. cache transfer between

Additionally, they manage and network-agnostic including storage—like range of networking

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Figure 7. NVIDIA Inference Transfer Library (NIXL) abstracts the complexity of data movement across heterogeneous memory and storage devices

NVIDIA Inference Transfer Library (NIXL) is a high-throughput, low-latency point-to-point communication library that provides a consistent data movement API to move data rapidly and asynchronously across different tiers of memory and storage using the same semantics. It is specifically optimized for inference data movement, supporting nonblocking and noncontiguous data transfers between various types of memory and storage.

NIXL enables NVIDIA Dynamo to interface with other communications libraries such as

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[NVIDIA Dynamo](#) builds on the successes [NVIDIA Triton](#), with a new modular architecture, distributed inference capabilities, and support for disaggregated serving, enabling it to deliver outstanding scaling performance in large scale multinode deployments.

AI Inference developers and researchers are invited to contribute to NVIDIA Dynamo on GitHub at [ai-dynamo/dynamo](#) and join the [new NVIDIA Dynamo Discord Server](#), the official NVIDIA server for developers and users of NVIDIA Dynamo. NVIDIA Dynamo is planned to be supported by [NVIDIA AI Enterprise](#) and available with [NVIDIA NIM](#) microservices for quick and easy deployment. NVIDIA AI Enterprise customers using Triton will continue to receive production branch support for their existing Triton deployments.

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About David Zier

David Zier is the director of Deep Learning System Software at NVIDIA. His organization is responsible for creating state-of-the-art AI inference serving software and tools, including Triton Inference Server, GenAI-Perf, and now NVIDIA Dynamo. David holds a PhD in Computer Engineering from Oregon State University with over 20 years of experience.

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About Kyle Kranen

Kyle Kranen is a deep learning software engineer at NVIDIA. He received a BS in electrical engineering and computer sciences from the University of California, Berkeley. At NVIDIA, his research is currently concentrated on recommender systems and time-series modeling.

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Inference Server and an AI solutions engineer. His focus is on enabling developers with a smooth transition from prototyping to high-

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Architecture. Omri has extensive experience in developing and integrating accelerated networking technologies used in hyperscale cloud. His focus is on software-hardware co-design and large-scale distributed systems. Omri holds a bachelor's degree in Electrical Engineering from the Technion - Israel Institute of Technology and a master's degree in Electrical Engineering from Tel Aviv University.

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April 17, 2025

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